Simulator Documentation

Paper: It's about Time — Cerf's Up, Communications of the ACM, April 2026

Date	2026-04-14
Model used	claude-sonnet-4-6 (Anthropic)
Simulation backend	event_process
Input files	3799690.pdf (PDF — 496 KB)
Parameters total	17 (12 extracted from paper, 5 assumed)
Scenarios	5
Figures planned	8
Dataset columns	19
Run ID	bf578196-7ad2-4dd3-8406-fde3493c4db8

Domain summary

This paper discusses integer overflow hazards in legacy timestamp representations, specifically the NTP 32-bit seconds wrap-around (February 6, 2036) and the UNIX 32-bit signed integer epoch overflow (January 19, 2038), and their potential impact on embedded, IoT, and legacy systems.

Backend rationale

The core phenomena are discrete overflow events triggered when a counter variable crosses a fixed threshold and wraps around to an erroneous value. An event-process backend is ideal because the dynamics are driven by counter accumulation over time with instantaneous state-change events at overflow boundaries, not continuous differential equations. The simulator must track multiple independent counters, detect wrap-around crossings, and log the resulting time-interpretation errors.

Value Proposition

This simulator bundle makes the abstract numerical arguments in Cerf's essay concrete and actionable. While the original essay describes the two overflow events in prose, the simulator produces time-series plots, event markers, and device-vulnerability heatmaps that allow engineers and policymakers to see exactly how large the misinterpretation errors are (on the order of 136 years, or ~4.3 billion seconds), when they occur, and how many devices remain at risk under different remediation strategies. The synthetic CSV data can be ingested into dashboards or compliance reporting tools, and the publication-quality figures are suitable for use in presentations, standards documents, and educational materials. Unlike reading the paper alone, the simulator allows users to interactively test their own patch-rate assumptions and initial legacy-device concentrations, providing a quantitative decision-support tool that the essay itself does not offer.

Who benefits:

- Embedded systems and IoT engineers responsible for long-lifecycle device firmware
- Security researchers and policy analysts assessing Y2K-analogous infrastructure risks
- Computer science educators teaching integer representation, overflow, and systems reliability

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How the generated data, figures, and analyses can be used:

• An IoT platform team can use the vulnerability heatmap figure (fig7) to determine the minimum annual patch deployment rate needed to reduce exposed device fraction below 5% before the 2036 NTP overflow, informing their release schedule and end-of-life planning.

• A university instructor can use the misinterpreted-year plots (fig3) and counter wrap-around figures (fig1, fig2) as interactive visual demonstrations of signed vs. unsigned integer overflow without requiring students to run real NTP or UNIX clocks.

• A standards body or industry consortium can use the scenario comparison curves (fig5) to argue for a coordinated patching campaign analogous to Y2K, backed by quantitative projections of residual device exposure under different industry response rates.

■ **Early Version Notice.** This is an early version of a reproducibility simulator generated automatically from scientific document(s). Validation results and data quality may vary depending on the source material, the complexity of the underlying model, and the completeness of the extracted parameters. Users are encouraged to review the generated *Simulator.py* code, verify outputs against the original paper, and improve or extend the simulator as appropriate. Contributions and corrections are welcome.

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1. Model Parameters

All parameters extracted from the paper or assumed as defaults.

Name	Value	Unit	Source	Description
ntp_max_counter	4294967295.0	seconds	extracted	Maximum value of 32-bit unsigned NTP seconds counter (2^32 - 1)
ntp_epoch_year	1900.0	year	extracted	NTP epoch start year: January 1, 1900
ntp_wraparound_year	2036.1	year	extracted	Calendar year of NTP 32-bit overflow: February 6, 2036
ntp_wraparound_seconds_from_epoch	4294967296.0	seconds	extracted	Seconds from Jan 1 1900 at which NTP 32-bit counter wraps: 136 years 4 days 6 hours 28 minutes 16 seconds
unix_max_counter	2147483647.0	seconds	extracted	Maximum value of 32-bit signed UNIX integer (2^31 - 1)
unix_epoch_year	1970.0	year	extracted	UNIX epoch start year: January 1, 1970
unix_wraparound_year	2038.05	year	extracted	Calendar year of UNIX 32-bit signed overflow: January 19, 2038, 03:14:07 UTC
unix_wraparound_seconds_from_epoch	2147483647.0	seconds	extracted	Seconds from Jan 1 1970 at which UNIX 32-bit counter hits max positive value
unix_post_overflow_interpreted_year	1901.89	year	extracted	Year that post-overflow UNIX counter is misinterpreted as: Dec 13, 1901 (approx 1901.95)
ntp_post_overflow_interpreted_year	1900.0	year	extracted	Year that post-overflow NTP counter is misinterpreted as without era correction: Jan 1, 1900
unix_64bit_max_years	292000000000.0	years	extracted	Duration before 64-bit UNIX counter overflows: approximately 292 billion years
patch_rate_per_year	0.05	fraction/year	assumed	Annual fraction of legacy/IoT devices upgraded from 32-bit to 64-bit time representation
initial_legacy_fraction	0.4	fraction	assumed	Fraction of deployed embedded/IoT systems still using 32-bit time at simulation start (year 2024)
seconds_per_year	31557600.0	seconds/year	extracted	Approximate seconds per Julian year (365.25 days)
simulation_start_year	2024.0	year	assumed	Start of the simulation timeline
simulation_end_year	2050.0	year	assumed	End of the simulation timeline to cover both overflow events with margin
time_step_days	30.0	days	assumed	Simulation time step in days (monthly resolution)

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2. State Variables

Name	Unit	Init Value	Range	Description
`ntp_seconds_counter`	seconds	0.0	[0.0, 4294967295.0]	32-bit unsigned integer counting seconds since Jan 1, 1900, used in NTP timestamps
`unix_signed_counter`	seconds	0.0	[-2147483648.0, 2147483647.0]	32-bit signed integer counting seconds since Jan 1, 1970 (UNIX epoch)
`unix_64bit_counter`	seconds	0.0	[0.0, 9.223372036854776e+18]	64-bit signed integer counting seconds since Jan 1, 1970, upgraded modern systems
`ntp_era`	a.u.	0.0	[0.0, 1000.0]	NTPv4 era number, incremented each time the 32-bit counter wraps around
`interpreted_unix_year`	year	1970.0	[1901.0, 2106.0]	Calendar year inferred from the 32-bit signed UNIX counter, goes wrong after overflow
`interpreted_ntp_year`	year	1900.0	[1900.0, 2106.0]	Calendar year inferred from the 32-bit NTP counter without era correction
`overflow_event_flag`	a.u.	0.0	[0.0, 1.0]	Binary flag: 1 when an overflow/wrap-around event has occurred in the current timestep
`time_error_seconds`	seconds	0.0	[0.0, 4294967295.0]	Absolute difference in seconds between the true current time and the misinterpreted timestamp after overflow
`legacy_device_fraction_affected`	fraction	1.0	[0.0, 1.0]	Fraction of legacy/IoT devices that have NOT been patched to 64-bit time representation
`simulation_year`	year	1970.0	[1970.0, 2100.0]	Current simulated calendar year in the simulator timeline

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3. Simulation Scenarios

Label	Description	Parameter Overrides
`no_patch_legacy_only`	All legacy 32-bit systems remain unpatched through 2050. Both NTP and UNIX overflows occur and misinterpretation errors are maximal. Represents worst-case IoT/embedded scenario.	patch_rate_per_year=0.0; initial_legacy_fraction=1.0
`slow_patch_baseline`	Realistic baseline: 5% annual patch rate starting in 2024, representing gradual enterprise and IoT upgrade cycle. Some devices still vulnerable at both 2036 and 2038 events.	patch_rate_per_year=0.05; initial_legacy_fraction=0.4
`aggressive_patch_campaign`	Aggressive patching campaign achieves 20% annual remediation rate, representing a coordinated industry response analogous to Y2K remediation efforts.	patch_rate_per_year=0.2; initial_legacy_fraction=0.4
`ntp_era_aware_systems`	Systems implement NTPv4 era awareness so the NTP 2036 overflow is handled gracefully, but UNIX 32-bit overflow remains unaddressed. Only the 2038 event causes misinterpretation.	patch_rate_per_year=0.05; initial_legacy_fraction=0.4
`high_legacy_slow_patch`	High legacy device concentration (70%) with very slow patch rate (2%), representing developing-region IoT deployments or long-lifecycle medical/industrial devices.	patch_rate_per_year=0.02; initial_legacy_fraction=0.7

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4. Output CSV Tables

Every CSV file produced by the simulator is documented below with full column listings and descriptive statistics.

4.1 sim_outputs/parameters_used.csv

Rows: 17 **Columns:** 5

#	Column Name	Type	Min / Examples	Max	Mean
1	`name`	text	ntp_max_counter, ntp_epoch_year, ntp_wraparound_year, ntp_wraparound_seconds_from_epoch, unix_max_counter	–	–
2	`value`	numeric	0.05	292000000000.0	17936263255.0288
3	`unit`	text	seconds, year	–	–
4	`source`	text	extracted	–	–
5	`description`	text	Maximum value of 32-bit unsigned NTP seconds counter (2^32 - 1), NTP epoch start year: January 1, 1900, Calendar year of NTP 32-bit overflow: February 6, 2036, Seconds from Jan 1 1900 at which NTP 32-bit counter wraps, Maximum value of 32-bit signed UNIX integer (2^31 - 1)	–	–

4.2 sim_outputs/scenario_summary.csv

Rows: 5 **Columns:** 51

#	Column Name	Type	Min / Examples	Max	Mean
1	`scenario_label`	text	no_patch_legacy_only, slow_patch_baseline, aggressive_patch_campaign, ntp_era_aware_systems, high_legacy_slow_patch	–	–
2	`patch_rate_per_year`	numeric	0.0	0.2	0.064

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#	Column Name	Type	Min / Examples	Max	Mean
3	initial_legacy_fraction	numeric	0.4	1.0	0.58
4	n_steps	numeric	318.0	318.0	318.0
5	ntp_overflow_steps	numeric	170.0	170.0	170.0
6	unix_overflow_steps	numeric	146.0	146.0	146.0
7	overflow_events_total	numeric	147.0	147.0	147.0
8	ntp_seconds_counter_mean	numeric	2027922700.8804	2027922700.8804	2027922700.8804
9	ntp_seconds_counter_std	numeric	1940470901.6784	1940470901.6784	1940470901.6784
10	ntp_seconds_counter_min	numeric	1791103.9999	1791103.9999	1791103.9999
11	ntp_seconds_counter_max	numeric	4294166399.9999	4294166399.9999	4294166399.9999
12	unix_signed_counter_mean	numeric	143039175.0535	143039175.0535	143039175.0535
13	unix_signed_counter_std	numeric	1938646869.9764	1938646869.9764	1938646869.9764
14	unix_signed_counter_min	numeric	-2145032897.0	-2145032897.0	-2145032897.0
15	unix_signed_counter_max	numeric	2147342399.0	2147342399.0	2147342399.0
16	unix_64bit_counter_mean	numeric	2114942399.0031	2114942399.0031	2114942399.0031
17	unix_64bit_counter_std	numeric	237941035.2449	237941035.2449	237941035.2449
18	unix_64bit_counter_min	numeric	1704110400.0	1704110400.0	1704110400.0
19	unix_64bit_counter_max	numeric	2525774399.0	2525774399.0	2525774399.0
20	ntp_era_mean	numeric	0.0	0.5346	0.1069
21	ntp_era_std	numeric	0.0	0.4988	0.0998
22	ntp_era_min	numeric	0.0	0.0	0.0
23	ntp_era_max	numeric	0.0	1.0	0.2
24	interpreted_ntp_year_mean	numeric	1964.261	1964.261	1964.261
25	interpreted_ntp_year_std	numeric	61.4898	61.4898	61.4898
26	interpreted_ntp_year_min	numeric	1900.0568	1900.0568	1900.0568

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#	Column Name	Type	Min / Examples	Max	Mean
27	interpreted_ntp_year_max	numeric	2036.0739	2036.0739	2036.0739
28	interpreted_unix_year_mean	numeric	1974.5326	1974.5326	1974.5326
29	interpreted_unix_year_std	numeric	61.432	61.432	61.432
30	interpreted_unix_year_min	numeric	1902.028	1902.028	1902.028
31	interpreted_unix_year_max	numeric	2038.0452	2038.0452	2038.0452
32	overflow_event_flag_mean	numeric	0.4623	0.4623	0.4623
33	overflow_event_flag_std	numeric	0.4986	0.4986	0.4986
34	overflow_event_flag_min	numeric	0.0	0.0	0.0
35	overflow_event_flag_max	numeric	1.0	1.0	1.0
36	time_error_ntp_seconds_mean	numeric	2296051698.5849	2296051698.5849	2296051698.5849
37	time_error_ntp_seconds_std	numeric	2142338337.0693	2142338337.0693	2142338337.0693
38	time_error_ntp_seconds_min	numeric	0.0	0.0	0.0
39	time_error_ntp_seconds_max	numeric	4294967295.0	4294967295.0	4294967295.0
40	time_error_unix_seconds_mean	numeric	1971903223.4906	1971903223.4906	1971903223.4906
41	time_error_unix_seconds_std	numeric	2140293795.4834	2140293795.4834	2140293795.4834
42	time_error_unix_seconds_min	numeric	0.0	0.0	0.0
43	time_error_unix_seconds_max	numeric	4294967295.0	4294967295.0	4294967295.0
44	legacy_device_fraction_affected_mean	numeric	0.0768	1.0	0.414
45	legacy_device_fraction_affected_std	numeric	0.0	0.0983	0.0694
46	legacy_device_fraction_affected_min	numeric	0.0022	1.0	0.3271
47	legacy_device_fraction_affected_max	numeric	0.4	1.0	0.58
48	legacy_frac_at_2036	numeric	0.0358	1.0	0.4046
49	legacy_frac_at_2038	numeric	0.0245	1.0	0.3904

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#	Column Name	Type	Min / Examples	Max	Mean
50	max_time_error_ntp_seconds	numeric	4294967295.0	4294967295.0	4294967295.0
51	max_time_error_unix_seconds	numeric	4294967295.0	4294967295.0	4294967295.0

4.3 sim_outputs/simulation_outputs.csv

Rows: 1,590 **Columns:** 19

#	Column Name	Type	Min / Examples	Max	Mean
1	simulation_year	numeric	2024.0	2050.037	2037.0185
2	scenario_label	text	no_patch_legacy_only	–	–
3	ntp_seconds_counter	numeric	1791103.9999	4294166399.9999	2027922700.8804
4	unix_signed_counter	numeric	-2145032897.0	2147342399.0	143039175.0535
5	unix_64bit_counter	numeric	1704110400.0	2525774399.0	2114942399.0031
6	ntp_era	numeric	0.0	1.0	0.1069
7	interpreted_ntp_year	numeric	1900.0568	2036.0739	1964.261
8	interpreted_unix_year	numeric	1902.028	2038.0452	1974.5326
9	interpreted_unix_year_64bit	numeric	2024.0	2050.037	2037.0185
10	overflow_event_flag	numeric	0.0	1.0	0.4623
11	ntp_overflow_occurred	numeric	0.0	1.0	0.5346
12	unix_overflow_occurred	numeric	0.0	1.0	0.4591
13	time_error_ntp_seconds	numeric	0.0	4294967295.0	2296051698.5849
14	time_error_unix_seconds	numeric	0.0	4294967295.0	1971903223.4906
15	legacy_device_fraction_affected	numeric	0.0022	1.0	0.414
16	patch_rate_per_year	numeric	0.0	0.2	0.064
17	initial_legacy_fraction	numeric	0.4	1.0	0.58
18	true_seconds_since_ntp_epoch	numeric	3913142400.0	4734806399.9998	4323974399.9999
19	true_seconds_since_unix_epoch	numeric	1704110400.0	2525774399.9998	2114942399.9999

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5. Output Figures

All PNG figures produced by the simulator, with thumbnails and descriptions from the simulation spec.

5.1 sim_outputs/fig1_ntp_counter.png

Filename	sim_outputs/fig1_ntp_counter.png
Title	NTP 32-bit Counter Accumulation and Wrap-Around (1900–2040)
Type	line
X axis	simulation_year
Y axis	ntp_seconds_counter
Inspired by	The NTP 32-bit seconds counter wraps around after 136 years on Feb 6, 2036, resetting interpretation to Jan 1, 1900

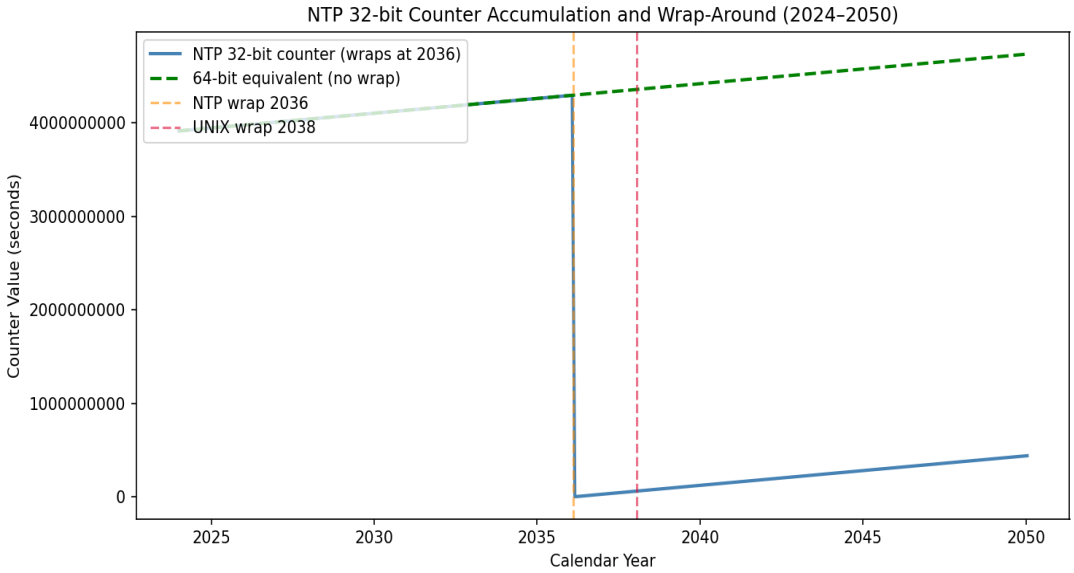

Figure explanation: This figure shows the NTP 32-bit unsigned integer counter value accumulating from zero (Jan 1, 1900) as a function of calendar year, reaching its maximum of 4,294,967,295 near February 2036 and then abruptly wrapping back to zero. The y-axis shows the raw counter value (0 to ~4.3 billion) and the x-axis shows calendar year. The reader should look for the sharp discontinuity at 2036 where the counter resets, and contrast this with a second line showing the equivalent 64-bit counter that continues smoothly without any visible wrap-around in this time range.

5.2 sim_outputs/fig2_unix_counter.png

Filename	sim_outputs/fig2_unix_counter.png
Title	UNIX 32-bit Signed Counter and Year 2038 Overflow
Type	line
X axis	simulation_year
Y axis	unix_signed_counter
Inspired by	UNIX 32-bit signed integer hits maximum positive value at 03:14:07 UTC on Jan 19, 2038, then goes negative

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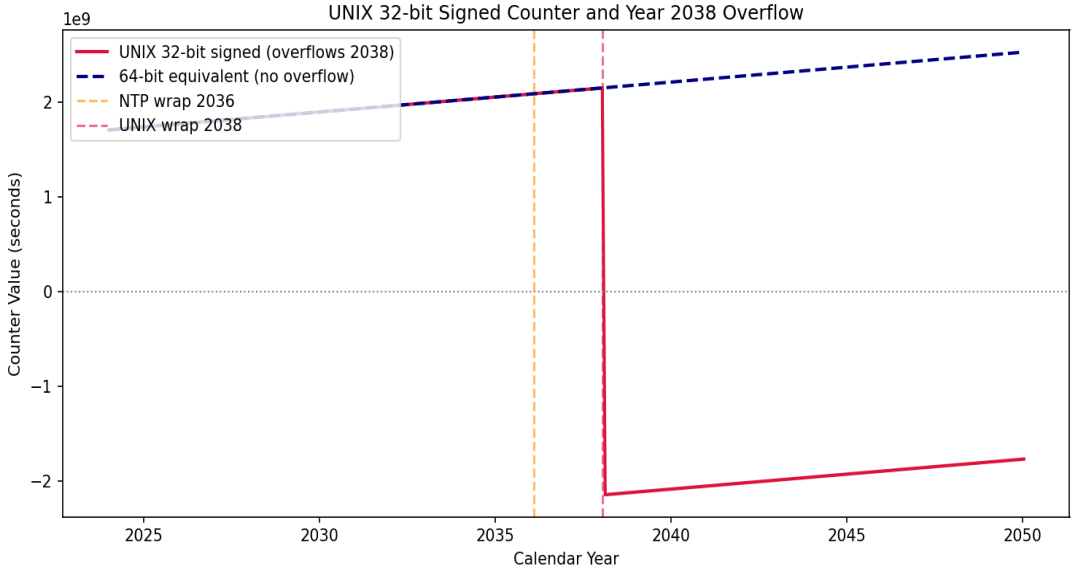

Figure explanation: This figure plots the UNIX 32-bit signed integer counter value from the UNIX epoch (1970) through 2050, showing it rising smoothly to its maximum of 2,147,483,647 in January 2038 and then flipping to large negative values (around -2,147,483,648) due to signed integer overflow. The x-axis is calendar year and the y-axis is the raw counter value. Two series are plotted: the broken 32-bit counter showing the catastrophic sign flip, and the well-behaved 64-bit counter for comparison. The visual impact of the sign reversal should be stark — a vertical drop from +2.1 billion to -2.1 billion.

5.3 sim_outputs/fig3_interpreted_unix_year.png

Filename	sim_outputs/fig3_interpreted_unix_year.png
Title	Misinterpreted Calendar Year After UNIX 2038 Overflow
Type	line
X axis	simulation_year
Y axis	interpreted_unix_year
Inspired by	After the 2038 overflow the system will think it is 08:45 pm on Dec 13, 1901

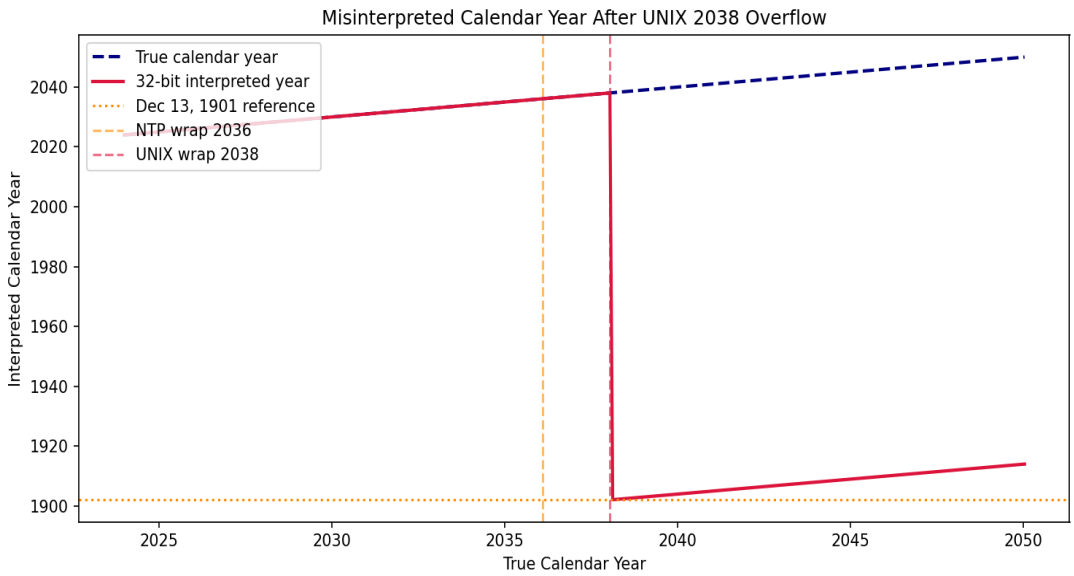

Figure explanation: This figure translates the raw counter value into the calendar year that a naive 32-bit UNIX system would report, from 1970 through 2050. Before overflow (pre-2038), the interpreted year matches the true year perfectly along the diagonal. After overflow in 2038, the 32-bit system suddenly reports a date in December 1901, creating a ~136-year retrospective error visible as a sudden plunge to the bottom of the y-axis. A horizontal dashed reference line marks 1901.95 (December 13, 1901). The magnitude of the time interpretation error — approximately 136 years — is the key takeaway.

5.4 sim_outputs/fig4_time_error.png

Filename	sim_outputs/fig4_time_error.png
Title	Time Interpretation Error (seconds) for 32-bit Systems Post-Overflow
Type	line
X axis	simulation_year
Y axis	time_error_seconds
Inspired by	Both NTP and UNIX overflows create large systematic time errors in legacy systems

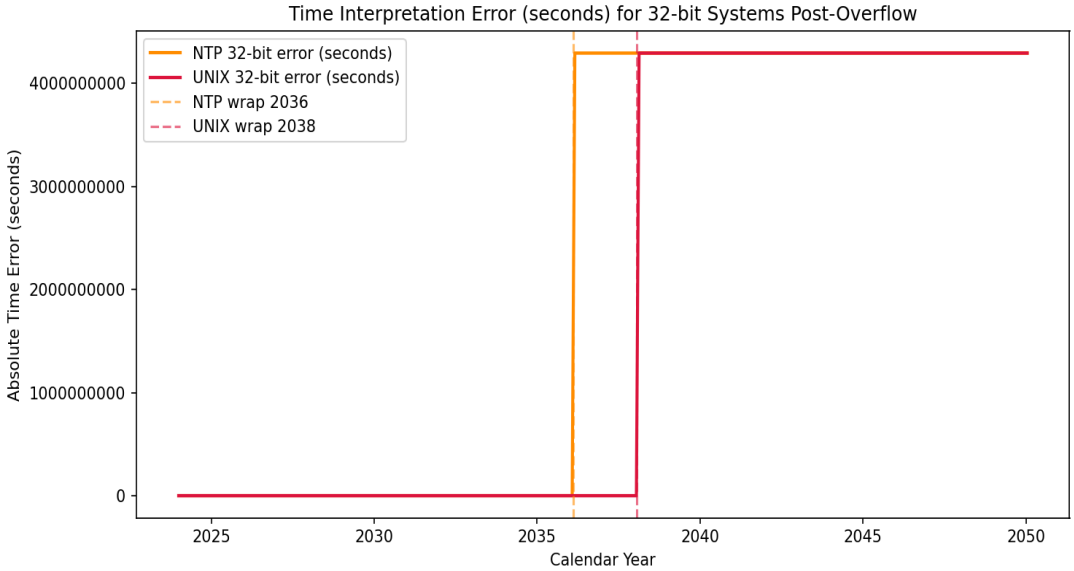

Figure explanation: This figure shows the absolute time interpretation error in seconds introduced by overflow, plotted against simulation year for both the NTP 2036 event and the UNIX 2038 event. Error is zero before each overflow event and jumps to a large fixed value immediately after (approximately 4.29 billion seconds for NTP reset to 1900, and approximately 4.29 billion seconds for UNIX rollback to 1901). Both overflow events appear as vertical step-function jumps, with a vertical dashed line marking each event year (2036 and 2038). The figure illustrates that without correction, the error persists indefinitely after the overflow.

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5.5 sim_outputs/fig5_legacy_fraction.png

Filename	sim_outputs/fig5_legacy_fraction.png
Title	Fraction of Unpatched Legacy/IoT Devices Under Different Patch Scenarios
Type	line
X axis	simulation_year
Y axis	legacy_device_fraction_affected
Inspired by	Legacy IoT, medical, and embedded devices not subject to regular updates remain at risk; analogy to Y2K remediation

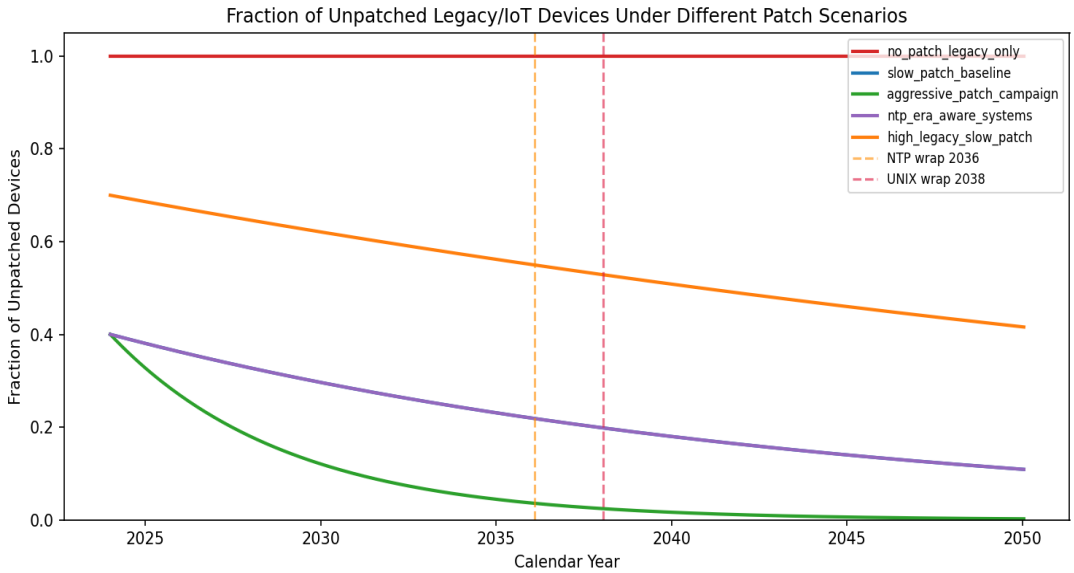

Figure explanation: This figure shows the fraction of deployed embedded and IoT devices still running 32-bit time representations (and therefore vulnerable to the 2036/2038 overflow events) as a function of calendar year, for all five scenarios. Each scenario is a separate colored line showing exponential decay of the unpatched fraction at different rates. Vertical dashed lines mark 2036 and 2038 to show what fraction of devices remain vulnerable at each overflow event. The reader can compare how aggressive patching (20%/year) achieves near-zero vulnerability before 2036, while slow or no-patch scenarios still have substantial exposed device populations at both dates.

5.6 sim_outputs/fig6_overflow_events.png

Filename	sim_outputs/fig6_overflow_events.png
Title	Overflow Event Timeline: NTP 2036 and UNIX 2038
Type	scatter
X axis	simulation_year
Y axis	overflow_event_flag
Inspired by	Two distinct overflow events: NTP on Feb 6 2036 and UNIX at 03:14:07 UTC Jan 19 2038

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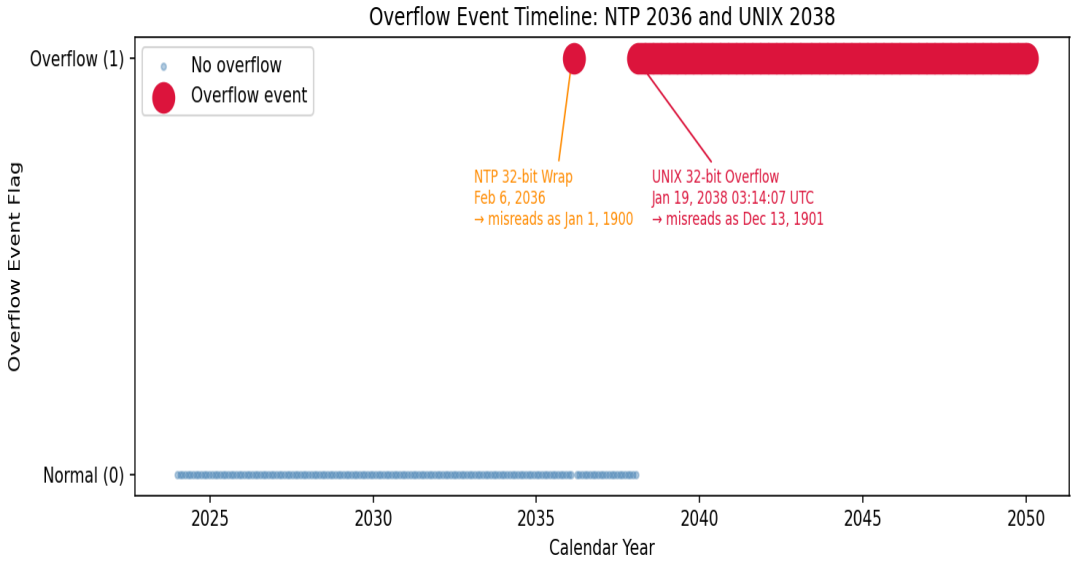

Figure explanation: This event timeline scatter plot marks the two critical overflow events as vertical markers on a year axis, with the overflow event flag (0 or 1) on the y-axis. NTP overflow at ~2036.1 and UNIX overflow at ~2038.05 are shown as tall vertical bars or large scatter points. Annotations label each event with the exact date, the type of counter that overflows, and the misinterpreted date the system would report. This figure serves as a compact reference card summarizing both critical dates, analogous to the direct statements in Cerf's essay.

5.7 sim_outputs/fig7_heatmap.png

Filename	sim_outputs/fig7_heatmap.png
Title	Vulnerability Heatmap: Legacy Device Exposure at 2036 and 2038 Events
Type	heatmap
X axis	simulation_year
Y axis	patch_rate_per_year
Inspired by	Devices not normally subject to regular software updates including IoT, medical, embedded sensors remain at risk

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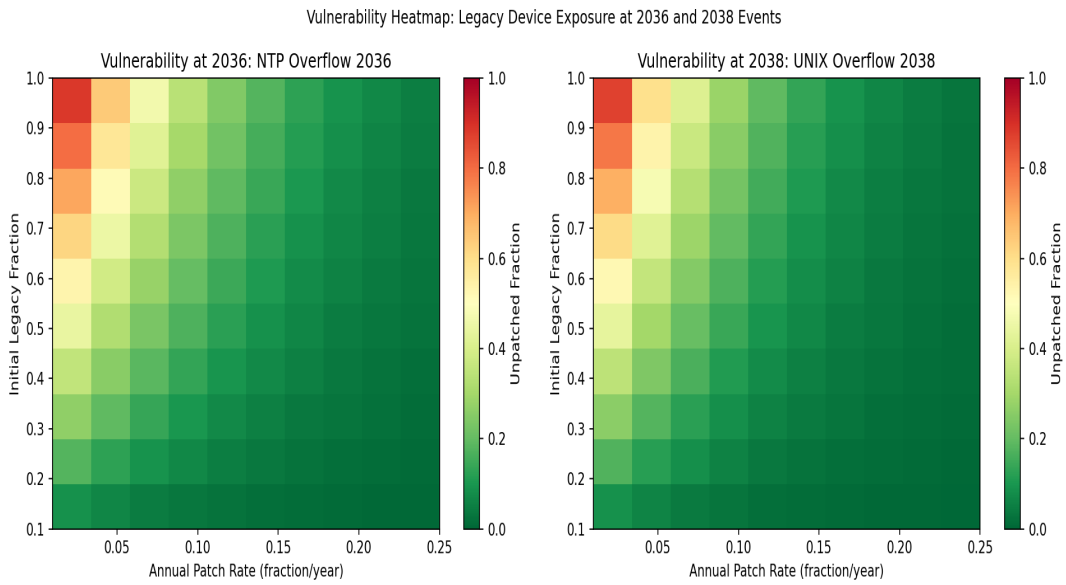

Figure explanation: This heatmap plots the fraction of devices still unpatched at the time of each overflow event (color intensity) as a function of initial legacy fraction (rows) and annual patch rate (columns). High initial legacy fraction combined with low patch rate (top-left) gives deep red indicating maximum exposure; low initial fraction with high patch rate (bottom-right) gives deep blue indicating near-zero exposure. The figure allows planners to read off at a glance what combination of initial conditions and remediation speed is needed to reduce vulnerability below a target threshold (e.g., 10%) before 2036 and 2038.

5.8 sim_outputs/fig8_ntp_era.png

Filename	sim_outputs/fig8_ntp_era.png
Title	NTP Era Number vs Calendar Year (Era-Aware vs Legacy Systems)
Type	line
X axis	simulation_year
Y axis	ntp_era
Inspired by	NTPv4 uses the concept of eras to deal with the 32-bit wrap-around; older systems may not implement era awareness

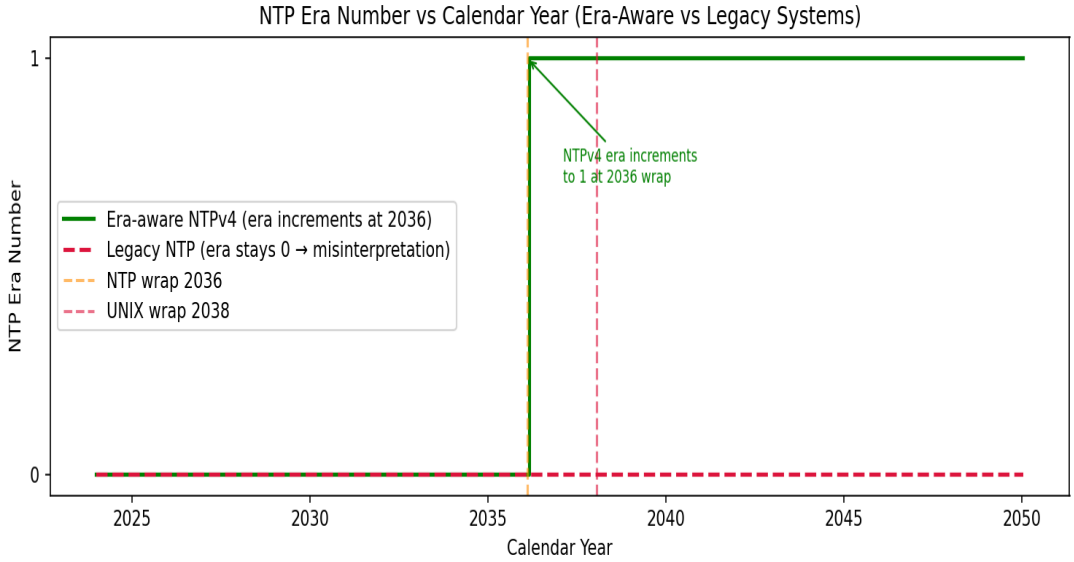

Figure explanation: This figure shows the NTPv4 era counter as a step function of calendar year, incrementing by 1 each time the 32-bit NTP counter wraps around (first wrap at ~2036, second wrap at ~2172, etc.). Era-aware systems correctly track the era and maintain monotonically increasing time. The figure plots two lines: one for era-aware NTPv4 systems (era increments at 2036) and one for legacy systems (era always stays at 0, leading to the 136-year misinterpretation). The step at 2036 is annotated to explain that era-aware systems remain correct while legacy systems diverge.

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